Sign Language Machine Translation A Review

By Suvarna Bhagwat

Chapter 14

Sign Language Machine Translation Systems: A Review

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ABSTRACT

For computer technology to make sense, it is imperative to have social relevance and the ability to also serve across sections of human society, including a chunk of physically deprived communities. The growth of IT applications in different Real-life domains boosted research in assistive technology for persons with a physical disability.

A lot more has been done to cater to the needs of visually impaired persons. But very less and only preliminary level IT-enabled technical aid is observed for persons with Hearing Impairment. Sign Language has been a prominent source of communication in this community for social connections. Again, the difference in modalities of Spoken and Sign Language has been one of the natural reasons for communication barriers between Hearing Impaired and the rest of society.

Advances in Machine Translation systems have triggered the idea of translation between Spoken Language and Sign Language. This article aims to cover Sign Language preliminaries and a global survey of Sign Language Machine Translation Systems with particular reference to Indian Sign Language. Different issues and aspects of Sign Language Machine Translation Systems such as translation methodology, grammatical features, domain, output interpretation, handling of Simultaneous morphological features are presented. Reported Sign Language Machine Translation Systems have been reviewed. After analyzing the gaps in reported systems, we have also proposed a tool to handle simultaneous morphological features of sign languages at run time.

KEYWORDS Assistive technology, Cross-modal Communication, Hearing Impairment, Machine Translation, Sign Language, Indian Sign Language

1.1 INTRODUCTION

The most important aspect of computer science is problem-solving, an essential skill for life. Since the evolution of computing mechanisms, various computational paradigms have been successively utilized to develop software and hardware for problem-solving. The technological and computational efforts taken in the initial period were more meant for scientific and commercial aids rather than real-life applications. An inclination towards the utilization of technology for the development of tools that can be employed in health, business, and social contexts has surely been observed at a later time. One of the significant areas where technology has been brought into being in the early 1990s is ‘Assistive Technology.’ It is a field where the design and development of equipment or product systems are done to increase, maintain, or improve the functional capabilities of persons with disabilities.

Persons with disabilities form a countable portion of the population. In a scene where every distinguished individual in society seeks acceptance and provision of equal level of opportunities, physically deprived people are not any exception. Assistive technology can aid towards special needs of such people and help them to raise their capabilities to the level of ordinary people. Since the establishment, this kind of technological aids can be found in the following abstract categories [1].

- Mobility impairments aid: Wheelchairs, Transfer devices, Walkers, Prosthesis
- Visual impairments aid: Screen readers, Braille embossers, Refreshable Braille display, Desktop video magnifier, Screen
- Magnification software, Large-print, and tactile keyboards, Navigation Assistance, Wearable Technology
- Hearing impairments aids: Hearing aids, Infrared listening devices, Amplified telephone equipment, speech-generating devices.

So as listed, in the near of beginning, assistive equipment was more of electronic and mechanical kinds of devices. Available electronic, mechanical devices are useful in increasing mobility capacity, amplification of visual and audio signals, etc. But in recent years, IT-enabled assistive products to be used in the field of education, sports, computer accessibility, memory aids, home automation are also being investigated.

One significant difference between hearing-impaired persons and persons with other kinds of disabilities is that the people in the latter category hardly face any kind of communication problem. Apart from not being able to hear and eventually not being able to speak, the major challenge faced by the hearing impaired is the communication barrier between them and the rest of society. As we know, communication requires a common medium of Language, signs, or behavior to exchange information between individuals. Communication can become a trouble if the medium used by communicators differ. In the case of two different spoken languages, bilingual dictionaries and basic knowledge of each other's Language can come to the rescue. But the trouble in communication may intensify if communicators are using languages of different modalities, viz. spoken language (Oral-auditory mode) and sign language (Gestural-visual mode).

Many times, Hearing Impaired persons are encouraged to be able to make use of Lip-reading. It can aid only up to a certain extent, as he/she should have a good grasp of spoken Language. Also, keen observation of slow and signer’s clean articulation can aid a hearing person to overcome the barrier. Still, this is also inadequate, as Sign Languages are as complex as spoken languages, also they do not confide in spoken Language.

Sign Language Interpreters can be utilized to make a way through this communication barrier. They can provide intermediary service to both ends. But several practical issues can limit the use of Sign Language interpreters now and then. Some of these issues can be –

- Low availability of Sign Language interpreters
- High hiring cost.
- Futile for confidential communication.
- Impractical for short communication e.g., shopping, booking, etc.
- Not viable for the reading of documents, newspapers, websites daily.
- Physical and time-bound constraints of Sign Language interpreters for long conversations

The use of computer-aided technology to lessen the language barrier was pioneered in the 1960s. 'Machine Translation,' which is the computer-aided translation from one Spoken Language to others, has been significantly employed for bridging the communication gap between spoken languages. Various approaches from rule-based, data-driven to machine learning have been successfully developed for a variety of spoken languages by researchers all over the world since then.

After the boost in linguistic research of Sign Languages, Machine Translation is also being utilized for reducing the barrier between Spoken Language and Sign Language. The first vital attempt for translation of Spoken Language to Sign Language was made in the 1990s using a Rule-based approach. It has been a thriving research area since then. This variation of Machine Translation is known as 'Sign Language Machine Translation.' It inputs spoken language text or speech and translates them to sign language. The output sign language can be represented in fron of a gloss or visual interpretation. Figure 1 gives an outlook of a typical Sign Language Machine Translation system.

Existing Sign Language Machine Translation systems translate Spoken Language text/speech to Sign Language GLOSS/articulation. Here, source and target languages are of different modalities, so correlated challenges can be seen.

This survey paper is arranged in the following way. Section-1.2 briefs about Sign language, along with its essential characteristics, which make Sign Language Machine Translation a challenging task. It also gives a concise survey of available writing systems for Sign Language. Section-1.3 surveys important Sign Language Machine Translation systems for Foreign Sign Languages. Section-1.4 specifically presents ventures carried out for Indian Sign Language. Gaps analysis is discussed in Section-1.5.

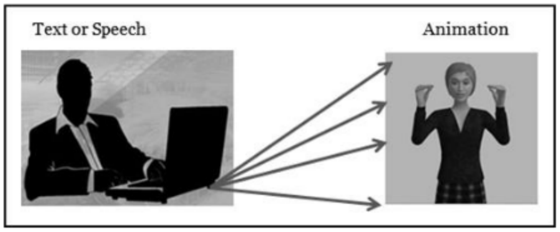

FIGURE 1 A Typical Sign Language Machine Translation System

1.2 SIGN LANGUAGE

From the literature survey, it has been observed that realization of Sign Language Machine Translation systems had been done quite late in the 1990s as compared to that of Machine Translation systems since the 1960s. It might be because of the late venture of sign languages' linguistic studies. One can surely say that Sign Languages are natural languages, as they have been evolved over a period of time just like any spoken language. Sign languages do have their grammatical structure, and Sign Language Machine Translation systems have to consider them.

In Sign Language, information is conveyed through the meaningful articulation of hands, arms, head, and shoulders. Such an articulation, containing a unique composition of Hand-shape, Palm-orientation, location of hands, and the movements form a manual component of a sign. Facial expressions like Eye-gaze, Raised-eyebrows, Puffed-Cheeks also play an essential role. They form a non-manual element of a sign. One sign may represent a single word, single phrasal structure, or even a small sentence in Spoken Language. Sign languages exhibit grammatical characteristics at all linguistic levels from phonetics, phonology, morphology, syntax, and semantics to even pragmatics.

Similar to spoken languages, there is no universal sign language, and almost every country has got its own official sign language. Through a literature survey, it has been observed that American Sign Language is apparently the most studied and well-documented Sign Language. Linguistic studies on Indian Sign Language were started around 1978 by Vasishta. Ulrik Zeshan does remarkable on field research and linguistic documentation in 2004 [2]. Though Indian sign language is being used by HI as well as interpreters, its linguistics study is gaining momentum in the last decade [3]. Presently online Video-based dictionary of words used in daily life is developed by Ramakrishna Mission Vivekananda Educational and Research Institute, Coimbatore, in collaboration with C-DAC, Hyderabad [4].

1.2.1 Challenges of Sign Language Machine Translation

Although both Spoken Language and Sign Language are natural languages, there are some interesting characteristics of Sign Language, which make them challenging for machine translation.

Basically, sign languages make use of manual gestures (hand movements), non-manual features (facial expressions), and the use of space simultaneously. This results in simultaneous nature of morphology. In this section, such features are aligned against that of Spoken Language [5].

- ***Simultaneity in articulation***

In the acoustic channel, it is hard to hear more than one thing at a time. Also, we only have one vocal track, so spoken speech is essentially linear. So, for Spoken Language, there can be just one serial stream of phonemes. Thus, spoken words are said to be sequential at the level of articulation. On the other side, in the case of sign languages, the visual system can perceive many things at once also multiple visible articulators: (two hands, head, shoulders, and facial expressions) can articulate at a time. Thus, in the case of Sign Language, multiple things can be articulated & perceived at a time. So, simultaneity is observed at the articulation level.

- ***Non-Manual Features***

As already stated, non-manual features like facial expressions, eye gaze, head movements play an important role in Sign Language grammar. They are used to express emotions, manner of the act. They are also used as morphological and syntactical markers to express interrogative sentences, negations, imperative sentences. It is again interesting to note that non-manual features are used simultaneously along with manual features.

- ***Signing space***

Generally, signs are articulated within the area in front of the sign's upper body. Space is also of prominent grammatical importance in Sign Languages. Signer's backside and front side space are associated with past and future tenses. Space is used to denote pronouns. Space is also used to build up location points, also known as anaphoric reference points and verb agreements.

- ***Morphological Incorporation***

In the literature of sign languages, the process of 'Incorporation,' is a way of combining two signs. In this process, a phonetic parameter of one sign is replaced by that of another sign. It results in a complex sign. Sign languages exhibit various kinds of incorporation, like numeral incorporation, classifier incorporation, and Ad-position incorporation.

- ***Writing/representation systems***

There are currently 7,097 living spoken languages in the world. Out of them, about 3,900 languages are unwritten. i.e., they don't have their writing systems, or they are unwritten.

On the other hand, one may think that sign languages cannot be written or read. But writing systems sign languages are available. The more interesting thing about them is that they are language-independent. As they actually describe the articulation done while signing, one can write any sign language using any of the writing systems. During the literature survey, four prominent writing systems are observed viz. GLOSS, STOKOE notation, SignWriting, and Hamburg Notation System (HamNoSys). More on these writing systems are discussed in the next subsection.

As shown in figure 1, the output of Sign Language Machine Translation systems is sign language in written and/or visual form. Choosing an appropriate writing system for representation of the target Sign Language is an eventual and important step in the development of Sign Language Machine Translation.

3.2.2 Sign Language Writing Systems

Articulation and perception of sign languages are Visual-spatial in nature. The information that they convey cannot be written and read as that of spoken languages. It took time to accept sign language as a formal language, even writing systems for sign languages were lately recognized. This section attempts a survey of existing sign language writing

- ***Annotation Systems.***

One can view it as one- or two-word translation in one Language for word or morpheme in another language. These systems transcribe the information observed from sign articulation. Sign Language linguistic knowledge is necessary for writing or reading annotations. In literature, such systems are named 'GLOSS.' It is generally used by linguists to represent signing sequence along with grammatical, spatial, and non-manual features (e.g., tenses, emotions, types of sentences, manner, etc.).

Spoken language stems are normally used to gloss over what is being articulated and gestured. Clearly, such a system can be employed for computational experimentation of sign language, as spoken language scripts can be used to write annotations.

GLOSS does not have universal standards. A transcriber decides the level of detail at which the Sign Language articulations can be described. It just portrays the signing sequence, but does not represent how actually signing will look like. Because of these shortcomings, this kind of system generally cannot be used for communication purposes but used for transcription [5].

- ***Pictorial Systems***

As discussed, annotation systems reveal morphological and syntactical features of sign language rather than phonological aspects. A pictorial scripting system, like SignWriting, SiS5 can be used to represent phonological level characteristics of signs. Such systems use compact icons having simple drawing elements like a circle, rectangle, lines, etc. to represent articulation as well as facial gestures [6].

- ***Symbolic Systems***

These are text-based systems for which ASCII fonts are available. So, they can be used for computational purposes. Some noticeable systems observed in this category are STOKOE, SignFont, SLIPA, ASL Ortho, and HamNoSys [7].

The available sign writing systems are studied based on various features like, Font type, Universality, Linearity, machine compatibility etc. Table 1 depicts a comparative study of the studied Sign Language writing systems against various characteristics.

TABLE 1 Comparison of Imminent Sign Language Writing Systems

Writing System	Can it be used to represent all Sign Languages?	Support for representation of Facial Expressions	Font Type	Iconicity	Arrange-ment	Machine Compatibility
Stokoe	No	No	Custom	Non-Iconic	Linear	ASCII
SignWriting	Yes	Yes	Pictorial Clip-art	Iconic	Nonlinear	ASCII, Unicode
HamNoSys	Yes	Yes	Custom	Iconic	Linear	Unicode, Convertible to Animation
SignFont	No	Some	Custom	Non-iconic	Linear	ASCII
ASLphabet	No	No	Iconic, Custom	Iconic	Linear	Online interface only
ASL Ortho	No	No	Roman	Non-iconic	Linear	ASCII, Unicode
SiS5	No	Yes	Pictorial	Iconic	Nonlinear	Handwritten
SLIPA	Yes	Yes	Roman	Non-iconic	Linear	ASCII, Unicode
ASLSJ	No	Yes	Roman	Non-iconic	Linear	ASCII, Unicode
SignScript	No	Yes	Custom	Non-iconic	Linear	ASCII
Gloss	Yes	Yes	Roman	Non-iconic	Linear	ASCII, Unicode

1.3 OVERVIEW OF SIGN LANGUAGE MACHINE TRANSLATION AT THE GLOBAL LEVEL

Machine Translation systems research for spoken languages can be observed since the 1960s. But similar kinds of efforts for sign language endeavored in the 1990s. More or less 20 systems have been deployed in Sign Language Machine Translation. Some systems for sign language generation through graphical avatars are observed. Several types of research for sign language recognition also have been done. This section delineates earlier attempts in Machine Translation for sign language. Along with the demonstration of how systems in this paradigm have progressed, it also discusses their pros and cons. Wherever feasible these systems are analyzed for

- Source and target languages
- Translation methodology

- Set of grammatical features of languages
- Domain
- Format of data
- Amount of data
- System architecture
- Details of graphical avatar generation
- Evaluation methods & results
- Handling of simultaneous morphological features of sign languages

In addition, it is also demonstrated how, over time, systems within this paradigm have progressed, as well as the more recent move toward more empirical methodologies. Discussion for foreign sign languages and Indian sign language is done separately in two subsequent sections.

1.3.1 The Zardoz System

It was the very first prominent attempt at using Machine Translation approaches for the translation of spoken Language to Sign Language. It was carried out by Veale et al. (1998) as a multilingual system for the translation of English text to Japanese Sign Language, American Sign Language, and Irish Sign Language [8]. The system has used the interlingua approach and includes multiple task-based modules. PATR-based unification grammar is used to represent interlingua. It also presents the design of a complete generation of Sign Language & a detailed avatar animation phase. DCL: Doll Control Language was used for animation generation [9].

Although this system was theoretically strong, it was minimally implemented. Also, no evaluation process is noted.

1.3.2 Translation from English to American Sign Language by Machine (TEAM)

Zhao et al. 2000 employed the TEAM system using a syntactic Transfer-based approach. It was initially proposed for English to American Sign Language, later on, also implemented for South African Sign Language. It was supported by the US Air force, the Naval office of the USA. The system architecture was divided into two phases. In phase-1, Synchronous tree-adjoining grammars (STAGs) are used to create source and target language parse trees. The output of phase-1 is American Sign Language gloss, which is able to incorporate American Sign Language morphological and adverbial aspects. The output of Phase-1 is used for graphical Avatar-based animation generation during phase-2. Human modeling & simulation technology is used to generate animation using a 3D graphical avatar [10].

This system is not domain-specific, but only limited morphological aspects have been experimented with. The project was successful in incorporating emotive capabilities in animation. Evaluation not noted.

1.3.3 Visual Sign Language Broadcasting (ViSiCast)

Ian Marshall and Eva Sáfar have utilized the Semantic Transfer-based approach for translation of English text to British Sign Language. Link Parser was used to analyze an input English text, whereas Prolog declarative clause grammar rules were used to convert this linkage output into a Discourse Representation Structure (DRS), which can store proposition level information e.g., Tenses, aspect, etc. Sign language output was represented using Head-driven Phrase Structured Grammar along with HamNoSys notations. Eventually, 'Signing Gesture Markup Language (SiGML)' was used to generate animation [24][11].

It was a domain-specific system. Animated weather forecast reports, simple informative messages at the post office are successfully generated. It lacks functionality for non-manual features. Again, no evaluation is noted.

1.3.4 A Multi-path Architecture

None of the systems discussed till now have considered classifier predicates (CPs). It is a phenomenon in which signers use special hand movements to indicate the location and movement of invisible objects (representing entities under discussion) in space around their bodies. A system by Matt Huenerfauth, University of Pennsylvania, USA 2006, proposed an interlingua approach where the interlingua is 3-D visualization of the objects in an English sentence. It also incorporates transfer and direct approaches into a single system. The translation process can follow one of the multiple paths offered by the system. Viz. Interlingua, transfer, or direct translation [12].

Evaluation of the system was done by native signers in terms of grammatical correctness, the naturalness of movement, and understandability. The system was also evaluated against animations created by a signer articulating CPs wearing motion-capture suits.

1.3.5 Research by RWTH Aachen Group

Statistical Machine Translation approach for Sign Language Machine Translation was utilized by Jan Bungerothe at RWTH Aachen university group in 2006. The system was employed for the German-DGS language pair with translation both to and from Sign Languages. Initially, the system was developed for whether report domain and later adopted for the Airline travel system. The Phrase-based SMT system already developed at RWTH was used for this purpose [13].

Output was not animated, so no manual evaluation was needed. Automatic evaluation is carried out on the gloss output, where BLEU scores range between 0.17 and 0.22.

1.3.6 Project Web-sign

Web-sign project 2004 was carried out at the University of Tunis, Tunisia to develop tools that will be able to make information over the Web accessible for the deaf. Under this project, Mohamed Jemni and Mehrez Boulares attempted an Example-Based machine learning system associated with genetic algorithms for the development of a Web-based interpreter of Sign Language to convert English to American Sign Language & French Sign Language. This system initially divides paragraphs into sentences. Then genetic algorithms were used for detecting various similar sequences existing in the initial sentence. The system will recognize the sentence if the system already learns it. If the system doesn't recognize this sentence, then the proximity search is launched to get the better sentence. Fuzzy logic was used to determine the emotional interpretation of the sentence [14][15].

A tool developed by a peer (Oussama El Ghoul) at the University of Tunis was used to create animations. Sign Markup Language (SML) was developed for the same.

1.3.7 Machine Translation using Examples (MaTrEx)

An approach that combines EBMT, SMT, translation memory (TM), and Information retrieval (IR) technologies for improving the quality & scale of Marker-based EBMT was successfully implemented at Dublin City University. Sara Morrissey et al. 2008 employed the same here for Sign Language Machine Translation. In this system, aligned sentences, words, and chunks are input to SMT for which a small corpus of 595 sentences was utilized. A working model of the system was created using the Air Travel Information System (ATIS) corpus [16].

An automatic evaluation was done for annotations/GLOSS. The decrease was observed in 'Word Error Rate' and 'Position Independent Word Error Rate' because of chunking. A manual evaluation was done for animation considering understandability, fidelity, naturalness, etc.

1.3.8 Japanese to Japanese Sign Language (JSL) glosses using a pre-trained model

This project has used a huge bilingual corpus of 1,3000 sentences to pre-train the language model for developing a transformer-based Neural Machine Translation Model. It has experimented on a domain-free vocabulary of size 6,000 words. The bilingual corpus contains the morphological le 2features of JSL like nodding, pointing, classifiers inscribed. The system output is JSL gloss and it is evaluated using the BLEU score. However, the visual generation of sign language is not reported [26].

1.3.9 Sign Language Production using Generative Adversarial Networks

In spoken to sign language translation systems have generally used a graphical avatar or prerecorded videos for the generation of sign language. But this is the first approach that has not used any graphical avatar for continuous video generation. The Neural Machine Translation along with the motion graph is used for the translation of spoken language sentences to sign language gloss. Then the generative model is used to create realistic videos. The system is evaluated using the BLEU score and WER [27].

Table 2 summarizes the main features of Sign Language Machine Translation Systems for Foreign Sign Languages, which are already discussed in detail.

1.4 OVERVIEW OF SIGN LANGUAGE MACHINE TRANSLATION FOR INDIAN LANGUAGES

Very few attempts are noted in the area of research performed in the paradigm of Sign Language Machine Translation for Indian Sign Language. Mainly, they have employed Rule-based approaches because of the scarcity of Indian Sign Language linguistic resources. This section discusses these attempts.

1.4.1 INGIT: Limited Domain Formulaic Translation from Hindi Strings to Indian Sign Language

This has been the very first attempt at Indian Sign Language. Purushottam Kar et al. 2008, have employed Rule-based machine translation for Hindi strings to Indian Sign Language strings. The system takes input from the reservation clerk and translates it into Indian Sign Language. It is

TABLE 2 Sign Language Machine Translation Systems for Foreign Sign Languages

Project	Source – Target Language	Sign Representation	MT Approach	Remark related to the handling of Simultaneous Morphology handling
ZARDOZ System (1994)	English - American, Japanese Irish SL	Doll Control Language, Graphical display	Knowledge-based interlingua	Spatial Features and Non-Manual Features are handled.
TEAM project (2000)	English - American Sign Language	GLOSS / 3-D Human modeling & simulation	Syntax level transfer based	STAG Trees are used to embed non-manual features in Gloss
ViSi-CAST (2002)	English - British Sign Language	SiGML 2-D Animation	Semantic level transfer based	Simultaneous morphological features are handled using manual intervention
Multi-path (2004)	English - American Sign Language	3-D virtual reality Avatar (NLI) software	Hybrid: Interlingua, Direct & Transfer	Non-Manual Features are handled
RWTH Statistical Translation (2007)	German - German Sign language (DGS)	SiGML 2-D Animation	Phrase-based Statistical Translation	Not Reported
Web-Sign Project (2006)	English - American & French Sign Language	Text Adapted Sign Modeling Language	Statistical Translation	Non-Manual Features are generated using Time Place, subject, and action
MaTrEx (2008)	English - Dutch Sign Language	HamNoSys/ SiGML 2-D Animation	Hybrid of SMT & EBMT	Not Reported
Japanese to JSL using Pre-trained Model (2020)	Japanese to Japanese Sign Language	Gloss	Neural Machine Translation	Nodding, pointing, classifiers inscribed in Gloss.
Sign Production using GAN (2020)	English to British Sign Language	Gloss, Realistic Videos	NMT, Generative Adversarial Networks for video generation	Non-manual features are generated using motion Graph Technology

based on a hybrid formulaic approach. Here, Fluid Construction Grammar is used to implement the Formulaic grammar [17].

As shown, the output of the Schematization module forms a structured sequence of signs for considering the temporal, location, person queries. Ellipsis resolution module extends Fluid Construction Grammar (FCG). The generated Indian Sign Language string is interpreted using HamNoSys & SiGML.

1.4.2 Dictionary-based Translation tool for Indian Sign Language

Tirthankar Dasgupta (2008), has developed a dictionary tool for the conversion of Hindi/Bengali/English to Indian Sign Language. Dictionary is able to store Hindi/English words, their meaning (with the help of Word-Net), POS, and corresponding HamNoSys. A sentence is accepted as input by the tool. Part of Speech tagging is done with the help of nlp.stanford.edu/software/tagger.shtml tagger. This tool can be used to associate signs to the words, phrases, or sentences of a spoken language text. The sign associated with each word is composed of its related Part-of-speech and semantic senses. HamNoSys & SiGML are used for interpretation [18].

In the later phase of this venture, a prototype for English to Indian Sign Language translation is employed [19]. The system can handle morphological functionalities like discourse, directionality, and classifier predicates minimally. Only simple sentences in English can be utilized as input.

1.4.3 Indian Sign Language Corpus for the domain of Disaster Management

In a joint venture of C-DAC (Pune) and Ali Yavar Jung Institute of Hearing Impaired (Mumbai), a bilingual corpus for English and Indian Sign Language has been developed. Corpus contains around 4000 words of disaster domain and around 600 sentences [20]. Initially, all the sentences in English were signed by a single signer. The synthetic avatars are created using a trace around the original video. Challenges with respect to contiguity, non-manual signs, and co-articulation were addressed.

This system can be used only for giving disaster-related messages to Deaf people, cannot be utilized to learn Indian Sign Language by people who do not have any sign language background.

1.4.4 Indian Sign Language from Text

A team at Thapar University, Patiala, India has made this admirable attempt under principal investigator Dr. Prateek Bhatia. They have developed an online multilingual multimedia Indian Sign Language dictionary. This dictionary contains 2000 English and 3286 Hindi words along with their Corresponding HamNoSys Codes [21].

Table 3 summarizes the main features of Sign Language Machine Translation Systems for Indian Sign Languages, which are already discussed in detail.

1.5 GAP ANALYSIS

Though there are ample NLP applications available, the languages they can process is a small subset of existing languages used worldwide. Many languages are still in need of NLP tools and computationally feasible resources that can be utilized by existing approaches/applications. The circumstances are applicable to many regional languages in India. The rich heritage of the regional languages of India has provided a massive scope for research in computational linguistics.

TABLE 3 Sign Language Machine Translation Systems for Foreign Sign Languages

Project/ Author	Source – Target Language	Sign Notation / MT Approach	General Remark	Remark related to the handling of Simultaneous Morphology handling
INGIT (2008) Purushottam Kar, IIT Khargpur	Hindi - Indian Sign Language	HamNoSys, SiGML 2-D Animation /Rule based Approach	Domain-specific. Small corpus (230 utterances, 90 words)	Simultaneous Morphological features are not handled graphically.
Dictionary-based prototype system (2008) Tirthankar Dasgupta, IIT, Kanpur, India	English – Indian Sign Language	HamNoSys/ SiGML 2-D Animation Rule-based Approach	Sign-videos incur memory overhead.	Morphological functionalities are handled minimally.
Disaster management Corpus building (2014) C-DAC, Ali Yavar Jung Institute for Hearing Impaired, Mumbai.	Disaster management messages ISL	Videos Tracing Corpus Size: 600 sentences 4000 words	Domain-specific. It can be used only for giving disaster management-related messages to Hearing Impaired.	Not reported
Indian Sign Language from Text (2016) Dr. Pratik Kumar, Thapar University, Patiala.	Hindi, English Indian Sign Language	HamNoSys/ SiGML 2-D Animation Rule-based Approach	It is a User-friendly system for the translation of Hindi and English sentences. Stanford Parser is used for parsing source sentences.	Any handling of simultaneous Morphological features is not reported

It has been noted from the available systems in the literature that, most of the spoken to sign language machine translation systems are developed using rule-based approach. Lack of bilingual parallel corpora and other computational resources have led to very few efforts using statistical and knowledge-based approaches.

Also, to our observation, most of systems lack the handling of simultaneous morphological features during Machine Translation. The linguistic study of Indian Sign Language has confirmed the fact that it exhibits Simultaneous Morphology, like other sign languages. The literature of Sign Language Translation Systems has considered the fact that it is necessary to handle the intrinsic features like grammatical use of signing space, classifier incorporation, numeral incorporation, ad position incorporation, simultaneous use of non-manual features, during the generation of sign language. We also didn’t find any system for translation of Marathi to Indian Sign Language.

1.6 DISCUSSION ON DESIGNED PROTOTYPE AND PROPOSED ENHANCEMENT

We have reported a novel approach translating a Marathi sentence in text form, to Indian Sign Language Gloss and Animation in [28]. Figure 2 shows the system architecture.

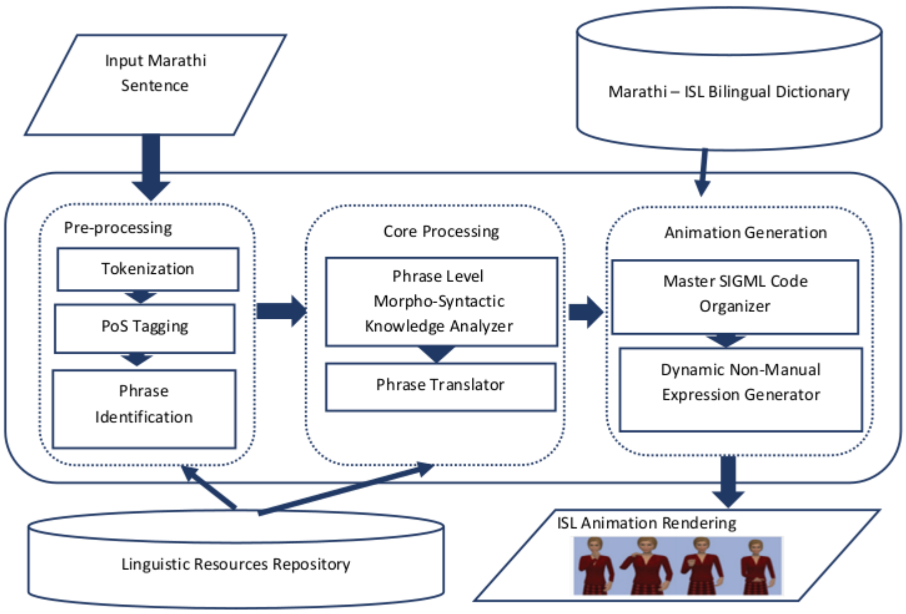

Figure 2 System architecture for Marathi Simple Sentence to Indian Sign Language Machine Translation System

The detailed morphological and syntactical comparative analysis of Marathi and Indian Sign Language has led us to practice the phrase-level rule-based approach for designing this system. The Marathi input sentence is preprocessed using tokenization, PoS Tagging and phrases are identified. The sentence is then translated to Indian Sign Language Gloss. The source and target language grammar rules are explicitly utilized in both the steps. As sign languages are visual languages, it was much necessary to visualize the output. We have used HamNoSys and SiGML to render the animation of generated gloss.

Now we are proposing enhancement to our prototype described in [28], by adding a new module for handling simultaneous morphological features at run time. The system architecture of the proposed extension is illustrated in figure 2.

As shown in the figure 3, a module, 'Dynamic Root sign Modifier', is needed to incorporate simultaneous morphological features in graphical animation. This module will gather the required information from the Machine Translation unit. It will make appropriate changes in the phonetic parameters of root signs stored in the database.

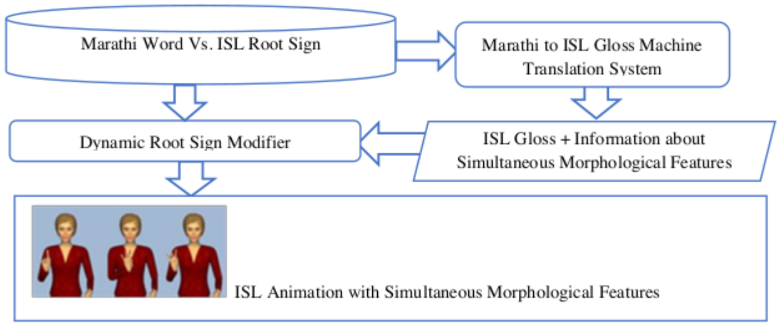

FIGURE 3 A proposal system to handle simultaneous morphology of Sign Languages during Machine Translation

1.7 CONCLUSION

This work has described the preliminaries of Machine Translation. It has discussed the basics of Sign Language linguistics in brief. We have studied and noted the challenges faced during spoken to sign language machine translation. Most of the challenges faced are due to simultaneous use of space, manual and non-manual gestures.

We have surveyed reported Sign Language Machine Translation systems for and presented a report on important systems. It observed that most of the sign languages all over the world are computational-resource scary languages. So mainly rule-based approaches are used for Machine Translation. Recently few systems have reported implementation of knowledge-based approaches using a comparable size of corpora.

As Marathi and Indian Sign Language lack compatible linguistic resources, we have developed a prototype system using a rule-based approach for translation. During the literature survey, we have observed that reported systems have not handled simultaneous morphological features of Indian sign language comprehensively. So, we find a need for a module that can uphold these features during machine translation. A brief idea of the proposed system is also discussed.

1.8 FUTURE SCOPE

We are looking forward to developing this system as a web-based application so that it can be availed in the process of teaching and learning of ISL. It can also be used as an online interpreter from Marathi to ISL.

Construction of Marathi Vs. Indian Sign Language bilingual corpora is the natural future proposal with main challenge of human expertise and extensive man hours. The availability of bilingual corpora would make it feasible to use statistical and knowledge-based approaches from machine translation.

Sign Language Machine Translation A Review

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